Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

A Freelancer Hub

The domain of the Project:

Full Stack Web Develpoment

Team Mentors (and their designation):

Prasanna Kundu-Software Engineer at Zaujivek

Team Members:

Mr. Gunti Srikanth

Period of the project

August 2025 to April 2026

Declaration

The project titled "Freelancer Hub" has been mentored by Prasanna Kundu, organised by SURE Trust, from August 2025 to April 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr.Gunti Srikanth
Signature : Gunti Srikanth

Mentor's Name : Prasanna Kundu
Designation: zaujivek

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

- The Freelancer Hub project is a full-stack web application designed to create a seamless digital platform connecting freelancers and clients. The primary objective of this project is to simplify the process of project posting, proposal submission, communication, and task management within a single integrated system.
- The application provides essential functionalities such as user authentication, project management, real-time messaging, and notification systems. By leveraging modern technologies like React.js, Node.js, MongoDB, and Firebase, the system ensures scalability, responsiveness, and real-time interaction between users.
- The key outcome of this project is the development of a fully functional freelancing platform that mimics real-world systems such as Upwork and Fiverr. The project demonstrates strong practical implementation of full-stack development concepts and highlights the ability to build industry-relevant applications.
- This project contributes significantly to enhancing employability skills by providing hands-on experience in designing, developing, and deploying a real-time web application.

Introduction

Background and Context

With the rapid growth of the digital economy, freelancing has become a major employment trend. Many professionals prefer flexible work environments, while organizations seek cost-effective solutions by hiring freelancers for short-term projects.

However, existing platforms often present challenges such as complex interfaces, high competition, and lack of transparency. There is a need for a simplified and efficient system that bridges the gap between freelancers and clients.

Problem Statement

Freelancers often face difficulty in finding genuine projects, while clients struggle to identify skilled professionals. Additionally, communication between both parties is often fragmented, leading to inefficiencies in project execution.

Scope of the Project

The scope of Freelancer Hub includes:

- User registration and authentication
- Project posting and management
- Proposal submission system
- Real-time messaging system
- Notification system for updates

Limitations

- Payment gateway integration is not fully implemented
- Advanced AI-based recommendations are not included
- Admin panel features are limited

Project Objectives

The primary objective of the **Freelancer Hub** project is to design and develop a full-stack web application that efficiently connects clients and freelancers on a unified platform. The system aims to simplify the process of project posting, proposal submission, communication, and project management through a user-friendly interface.

One of the key objectives of this project is to implement a secure authentication and authorization system using modern techniques such as JSON Web Tokens (JWT), ensuring that user data and system access remain protected. Additionally, the project focuses on enabling role-based functionality, where users can operate either as clients or freelancers with different capabilities and access levels.

Another important objective is to provide a structured workflow for project management. Clients can create and manage projects, while freelancers can explore available opportunities and submit proposals. This ensures a smooth interaction between both parties and improves overall efficiency.

Expected Outcomes and Deliverables

The expected outcome of this project is a fully functional freelancing platform that demonstrates the practical implementation of real-world web application features. The application should provide seamless user experience, efficient data handling, and reliable communication between users.

The key deliverables of the project include:

- A responsive frontend interface built using React.js and Tailwind CSS
- A robust backend developed using Node.js and Express.js
- A well-structured MongoDB database for storing application data
- Secure authentication and role-based access control
- A project and proposal management system
- Real-time chat functionality between users
- Push notification system integrated using Firebase
- Deployment of the application on a live platform

In addition to the technical deliverables, the project also provides hands-on experience in designing, developing, and deploying a complete full-stack application. It enhances problem-solving skills, improves understanding of system architecture, and prepares the developer for real-world industry challenges.

Methods / Technology Used

The Freelancer Hub project follows a **client–server architecture** using the MERN stack. The development approach was modular, where each feature (authentication, projects, proposals, messaging, notifications) was implemented as an independent module and then integrated.

- **Frontend (React.js + Vite + Tailwind CSS):**
  Built a responsive UI with component-based architecture. Routing handled via React Router. State managed using React hooks.
- **Backend (Node.js + Express.js):**
  Developed RESTful APIs for authentication, projects, proposals, messages, and notifications. Middleware is used for JWT-based route protection.
- **Database (MongoDB + Mongoose):**
  Designed collections for Users, Projects, Proposals, Messages, and Notifications. Relationships handled via ObjectId references.
- **Real-time Communication (Socket.IO):**
  Implemented bi-directional communication for instant messaging between users.
- **Push Notifications (Firebase Cloud Messaging):**
  Generated FCM tokens on login, stored in DB, and triggered notifications from backend using Firebase Admin SDK.
- **Media Handling (Cloudinary):**
  Used for uploading and storing user profile images.

Tools / Software Used

- **Development Tools:** VS Code, Git, GitHub
- **API Testing:** Postman
- **Database:** MongoDB Atlas
- **Deployment:** Vercel (Frontend), Render/Node server (Backend)
- **Libraries:** Axios, JWT, bcrypt, Socket.IO, Firebase SDK

Data Collection Approach

The project primarily uses **user-generated data** collected through forms:

- User registration data (name, email, role)
- Project details (title, description, budget)
- Proposal data (bid, cover message)

Project Architecture

The system follows a **three-tier architecture**:

- **Presentation Layer (Frontend):**
  - React UI handles user interactions
  - Sends API requests via Axios
- **Application Layer (Backend):**
  - Express server processes requests
  - Implements business logic and authentication
  - Integrates with Firebase and Socket.IO
- **Data Layer (Database):**
  - MongoDB stores all application data
  - Mongoose schemas define structure and relationships

Workflow:

- User logs in → JWT generated
- Client creates project → stored in DB
- Freelancer submits proposal → linked to project
- Chat via Socket.IO for communication
- Backend triggers Firebase notification
- Data updated and reflected in UI

Results

The project successfully achieved its intended objectives:

- A fully functional freelancing platform was developed
- Users can register, login, and manage profiles
- Clients can post projects and freelancers can submit proposals
- Real-time messaging between users is operational
- Push notifications are triggered for key events
- The application is deployed and accessible online

The system demonstrates efficient performance, smooth navigation, and proper integration between frontend and backend components.

Project GitHub Link

Frontend Repository: https://github.com/guntisrikanth0/SureTrust-freelancerfrontend.git
Backend Repository: https://github.com/guntisrikanth0/SureTrust-freelancerbackend.git

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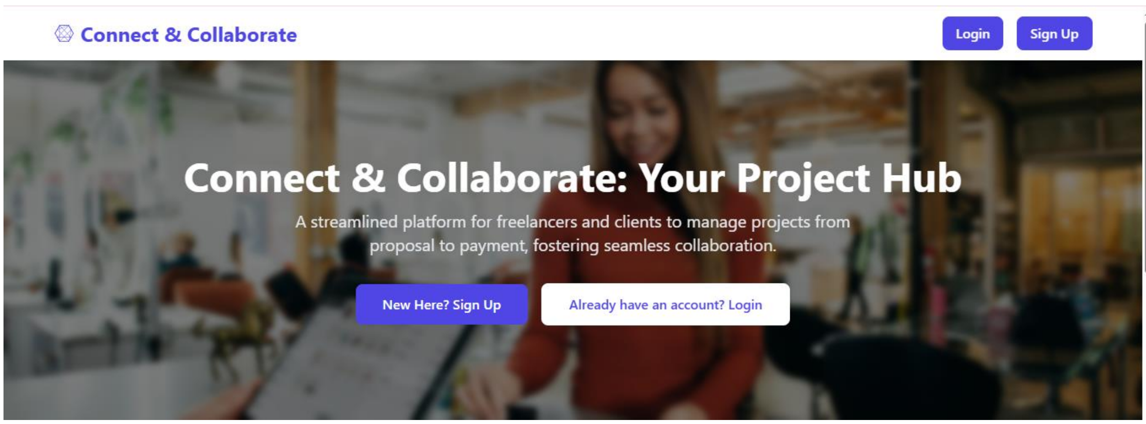
SCREENSHOT EXPLANATIONS

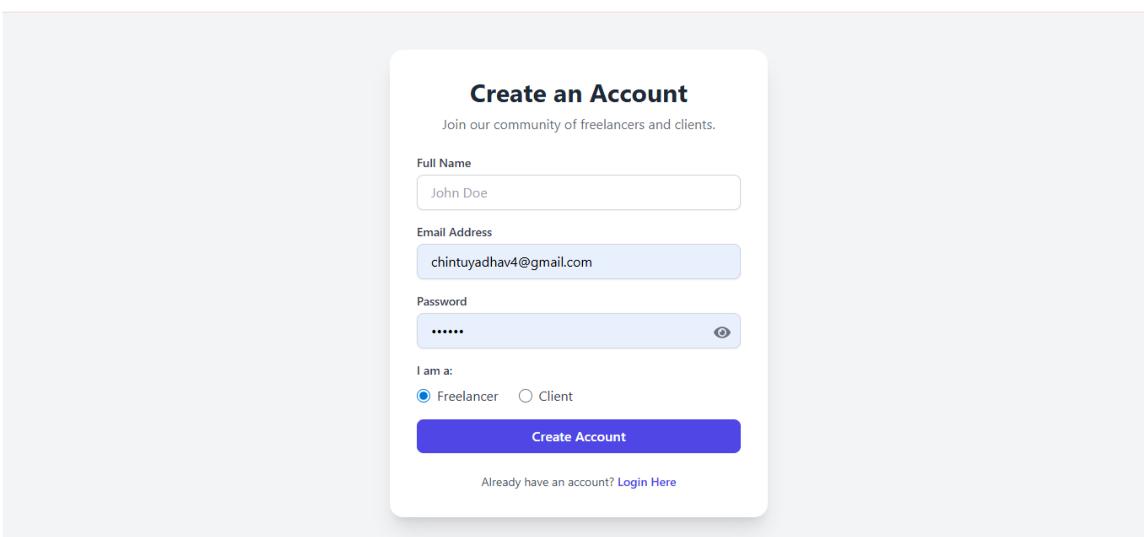
Figure 1: Landing Page of Freelancer Hub



Explanation:

The landing page provides an overview of the platform where users can understand the purpose of Freelancer Hub.Users can navigate to sign in or register as either a freelancer or a client. The design focuses on simplicity and user engagement.

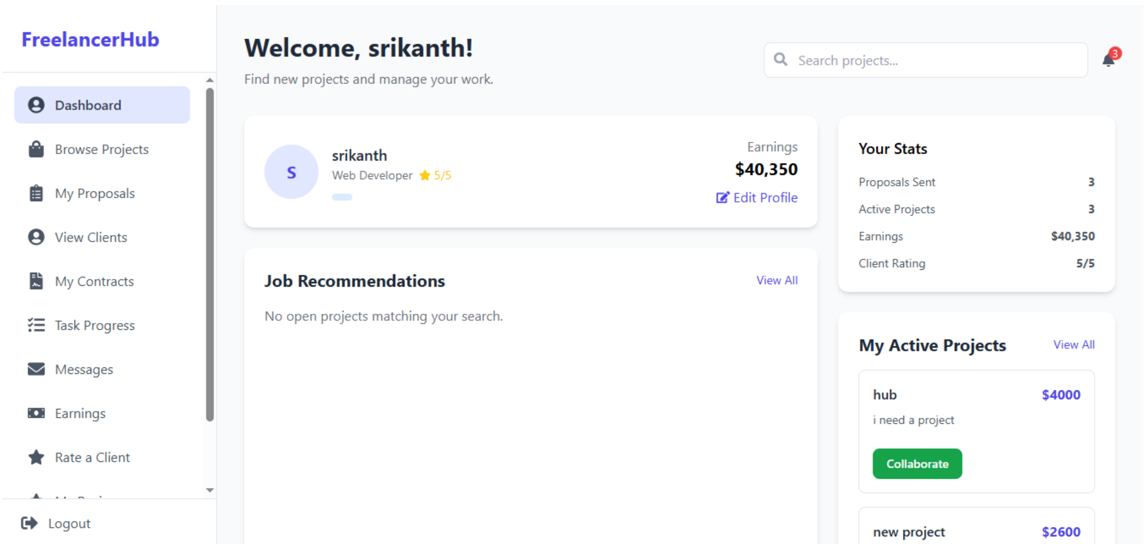
Figure 2: User Registration Interface



Explanation:

This page allows new users to create an account by entering their personal details such as name, email, password, and account type (client or freelancer).

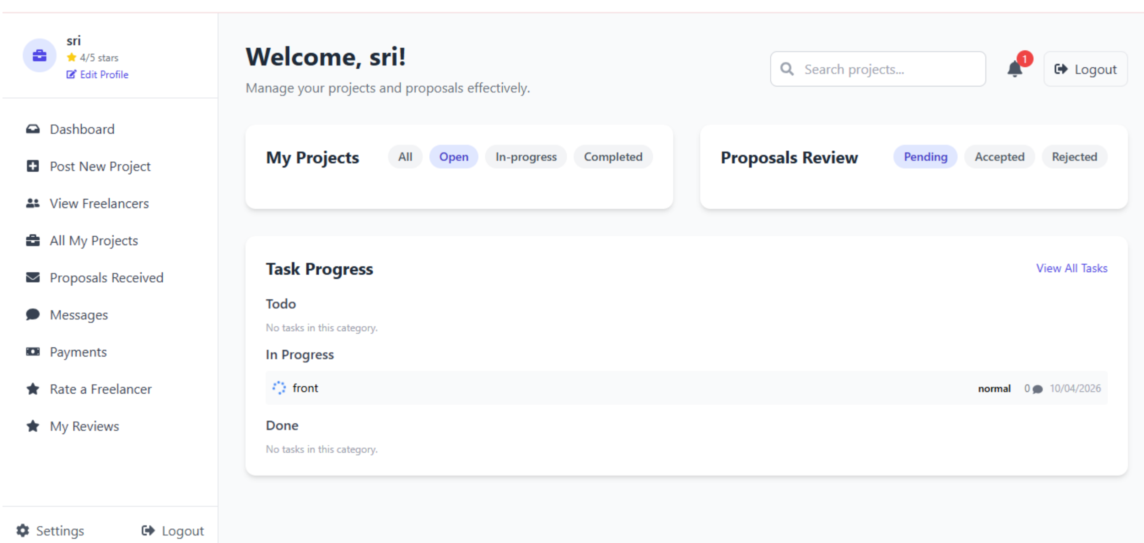
Figure 3: Freelancer Dashboard Overview



Explanation:

The freelancer dashboard provides a summary of the freelancer's activity, including active projects, completed projects, pending proposals, and total earnings. It helps freelancers track their work and performance in a structured manner.

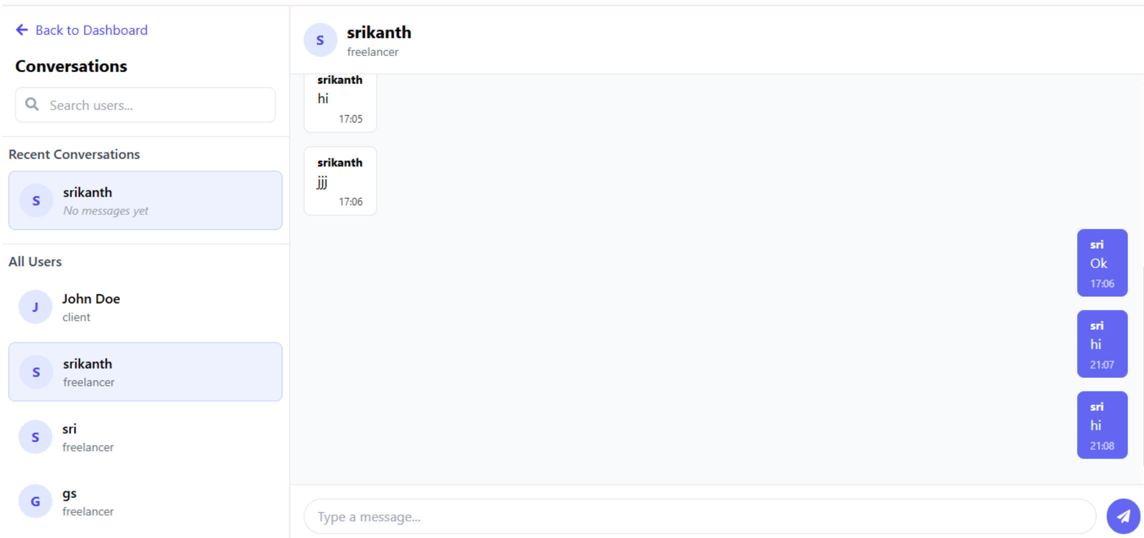
Figure 4: Client Dashboard Overview



Explanation:

The client dashboard allows users to manage their posted projects. It shows total projects, projects in progress, completed projects, and new proposals received. Clients can monitor project progress and interact with freelancers.

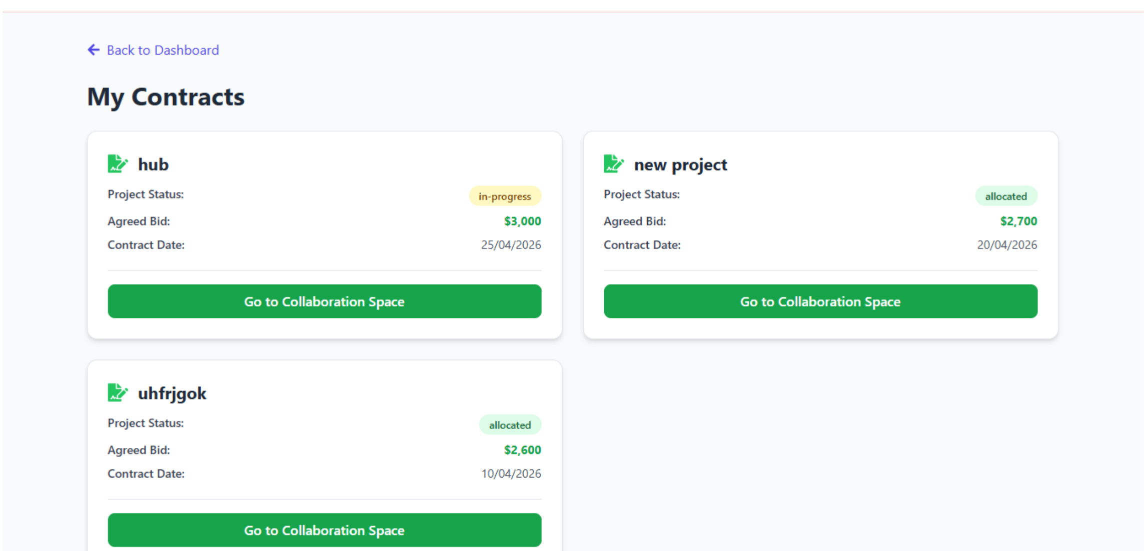
Figure 4: Real-Time Messaging Interface



Explanation:

The chat interface enables real-time communication between clients and freelancers. Messages are exchanged instantly using Socket.IO. This feature improves collaboration and clarity during project execution

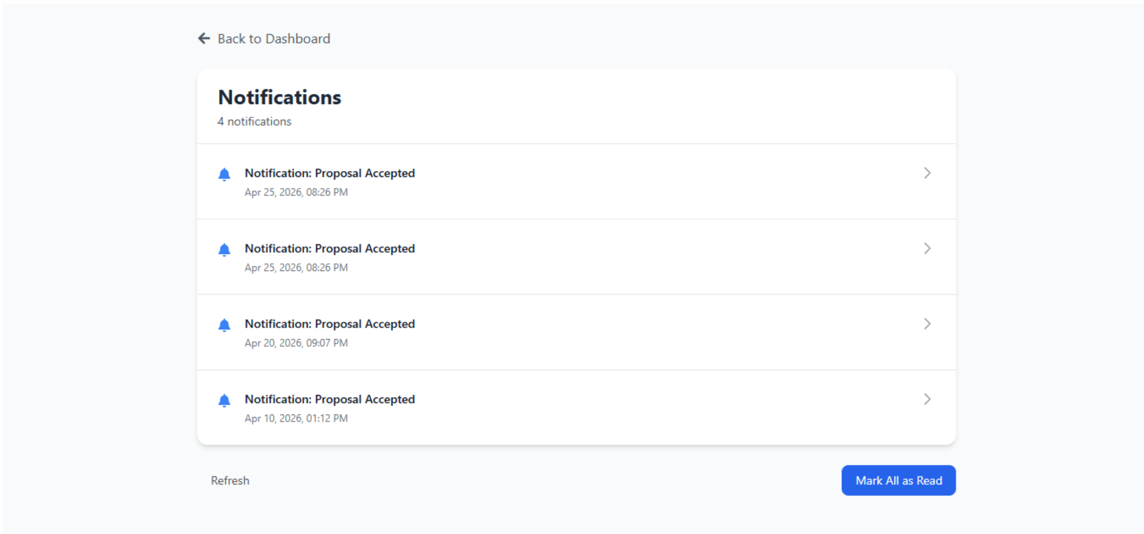
Figure 8: Freelancer contacts



Explanation:

This page displays the freelancer's earnings, pending payments, overdue payments, and total invoices. It provides a financial overview and helps freelancers track their income efficiently.

Figure 5: Notification Indicator



Explanation:

The notification system alerts users about important events such as new proposals or messages. Notifications are displayed both inside the application and through browser push notifications using Firebase.

Technical Learnings

During the development of the **Freelancer Hub** project, significant knowledge was gained in full-stack web development. The project provided hands-on experience in building a complete application using modern technologies.

One of the major learnings was working with the **MERN stack (MongoDB, Express.js, React.js, Node.js)**. Understanding how frontend and backend communicate through REST APIs helped in designing scalable and maintainable systems.

The implementation of **authentication using JWT** improved understanding of secure user management and protected routes. Additionally, working with **bcrypt** helped in learning secure password storage practices.

The project also involved **real-time communication using Socket.IO**, which provided insights into building interactive applications where data updates instantly without refreshing the page.

Another key learning was integrating **Firebase Cloud Messaging (FCM)** for push notifications. This helped in understanding how modern applications notify users in real time, both within the application and at the system level.

Working with **MongoDB** enhanced knowledge about NoSQL databases, schema design, and handling relationships using ObjectIds. The use of **Cloudinary** for image storage introduced the concept of cloud-based media management.

Problem-Solving and Debugging

Throughout the project, multiple challenges were faced, especially during integration between frontend and backend. Debugging API errors, handling asynchronous operations, and fixing real-time communication issues helped in developing strong problem-solving skills.

Understanding errors related to authentication, notification delivery, and deployment required patience and systematic debugging. These challenges improved the ability to analyze problems logically and implement effective solutions.

Project Management and Development Experience

The project provided valuable experience in managing a real-world application from development to deployment. It involved planning features, organizing code into modules, and maintaining a clean project structure.

Working on different modules such as authentication, project management, chat system, and notifications helped in understanding how large applications are built step by step. It also improved time management and the ability to prioritize tasks effectively.

Version control using Git and GitHub played an important role in tracking changes and maintaining code history. Deployment of the application provided exposure to real-world hosting environments.

Overall Experience

Overall, the development of Freelancer Hub was a highly enriching experience. It provided practical exposure to industry-relevant technologies and strengthened both technical and analytical skills.

The project helped in building confidence in developing full-stack applications independently. It also improved understanding of how real-world platforms function, from user interaction to backend processing and data management.

This experience has prepared the developer to handle real-world software development challenges and has created a strong foundation for future learning and career growth.

Future Scope

Although the current system is fully functional, there are several areas where the project can be further enhanced to make it more robust and industry-ready.

One of the major improvements would be the integration of a **secure payment gateway** (such as Stripe or Razorpay), allowing real-time transactions between clients and freelancers. This would make the platform more complete and practical for real-world usage.

Another important enhancement is the implementation of an **AI-based recommendation system** that can suggest relevant projects to freelancers based on their skills and past activity, and recommend suitable freelancers to clients.

Conclusion

The **Freelancer Hub** project was successfully designed and developed as a full-stack web application that connects clients and freelancers on a single platform. The primary objective of simplifying project management, proposal handling, communication, and notifications was effectively achieved.

The application enables clients to post projects and manage them efficiently, while freelancers can explore opportunities, submit proposals, and track their work progress. The integration of real-time chat functionality ensures seamless communication between users, improving collaboration and reducing delays.

The project also demonstrated the effective use of modern web technologies such as React, Node.js, Express, and MongoDB, along with third-party services like Cloudinary and Firebase. Overall, the project successfully met its objectives and provided a practical understanding of building scalable and interactive web applications.